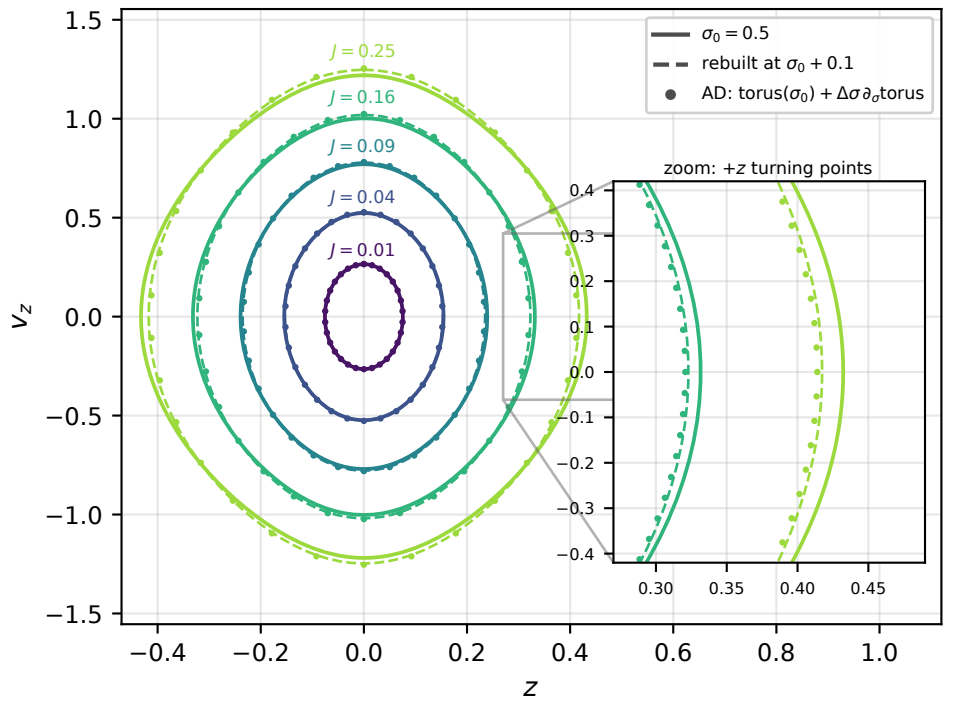
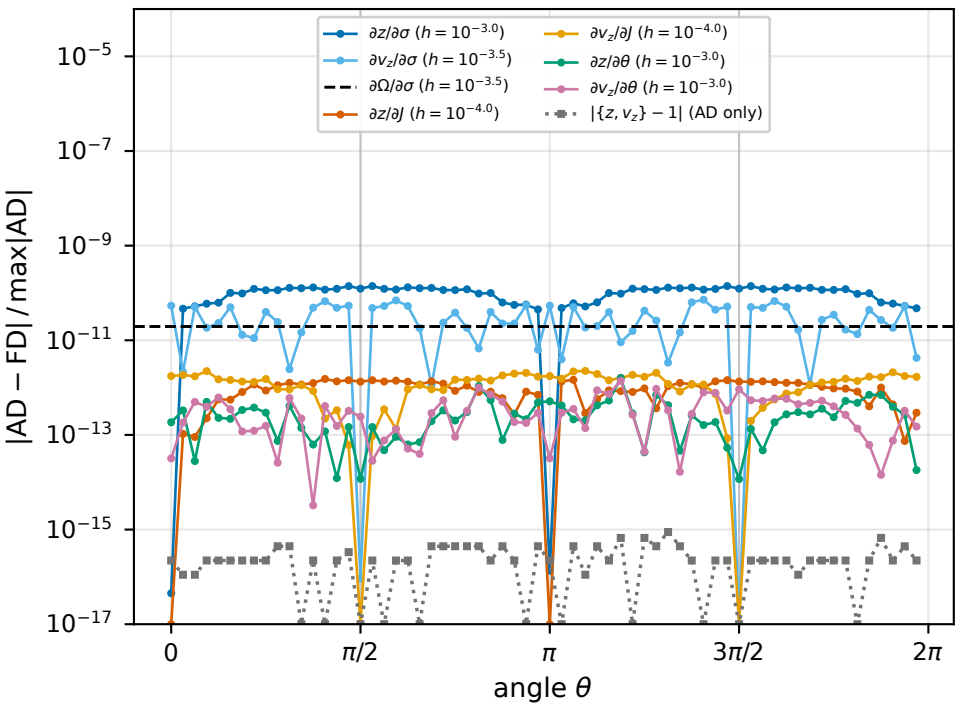
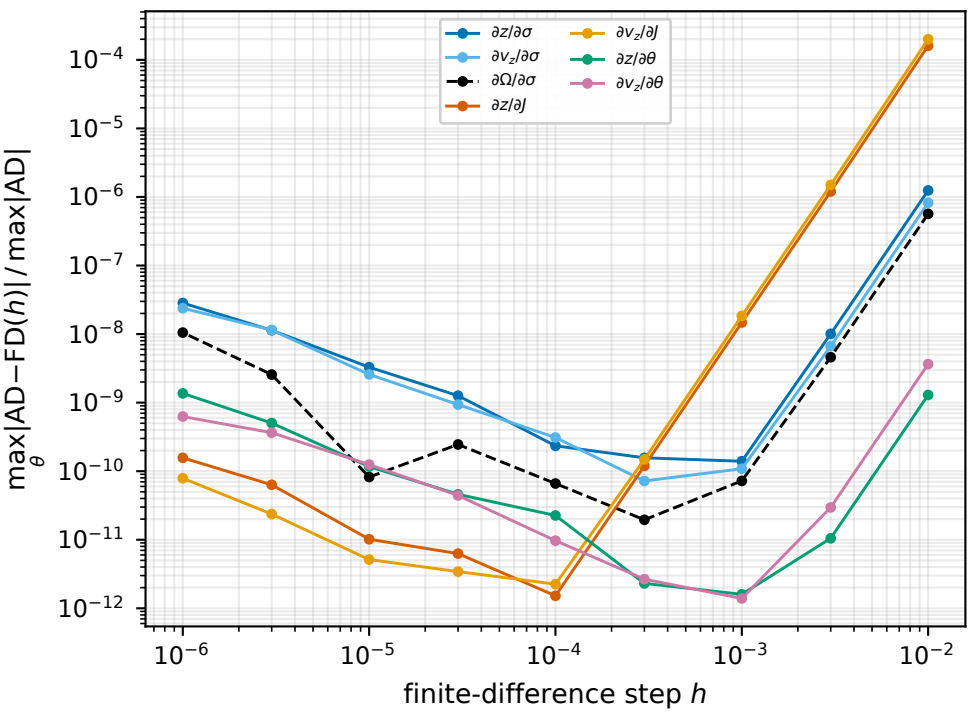
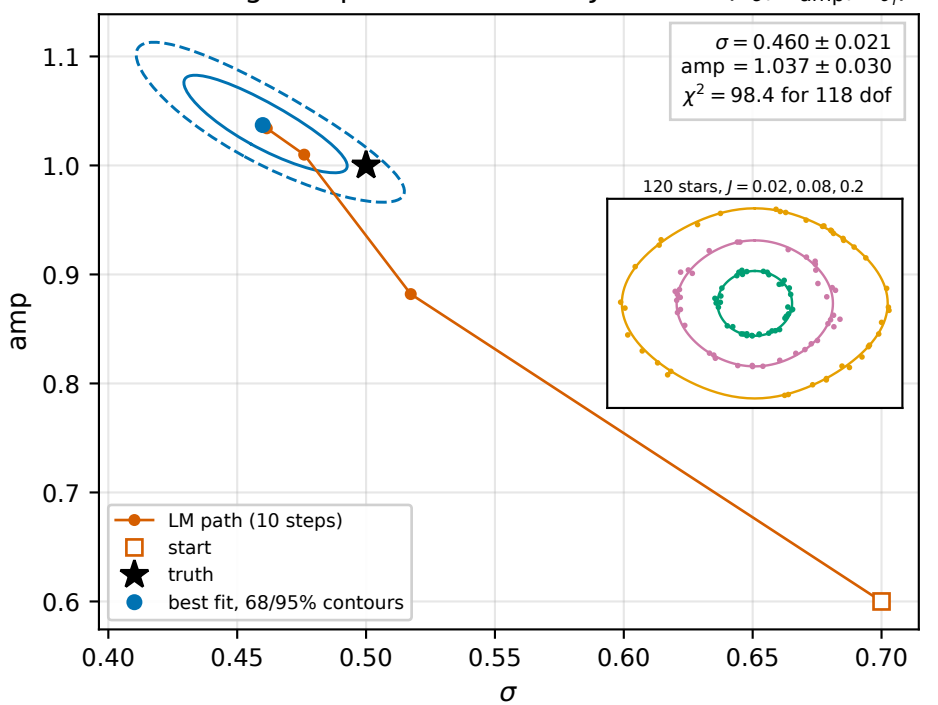

(a) isothermal-disk tori at fixed J and their AD sensitivity to σ ($\Delta\sigma = 0.2\,\sigma_0$)
$\sigma_0 = 0.5$
rebuilt at $\sigma_0 + 0.1$
AD: torus(σ_0) + $\Delta\sigma\,\partial_\sigma$torus
$J = 0.25$
$J = 0.16$
$J = 0.09$
$J = 0.04$
$J = 0.01$
zoom: $+z$ turning points
v_z
z


(b) AD vs 5-point central FD along the $J = 0.06$ torus (σ-FD rebuilds the family at $\sigma_0 \pm h$)
$\partial z/\partial\sigma$ ($h = 10^{-3.0}$)
$\partial v_z/\partial\sigma$ ($h = 10^{-3.5}$)
$\partial\Omega/\partial\sigma$ ($h = 10^{-3.5}$)
$\partial z/\partial J$ ($h = 10^{-4.0}$)
$\partial v_z/\partial J$ ($h = 10^{-4.0}$)
$\partial z/\partial\theta$ ($h = 10^{-3.0}$)
$\partial v_z/\partial\theta$ ($h = 10^{-3.0}$)
$|\{z, v_z\} - 1|$ (AD only)
$|\mathrm{AD} - \mathrm{FD}| / \max|\mathrm{AD}|$
angle θ


(c) the 5-point central FD converges onto AD ($\propto h^4$) down to its round-off floor
$\partial z/\partial\sigma$
$\partial v_z/\partial\sigma$
$\partial\Omega/\partial\sigma$
$\partial z/\partial J$
$\partial v_z/\partial J$
$\partial z/\partial\theta$
$\partial v_z/\partial\theta$
$\max_\theta|\mathrm{AD}-\mathrm{FD}(h)| / \max|\mathrm{AD}|$
finite-difference step h


(d) fitting the disk from noisy (z, v_z) on 3 tori: Levenberg-Marquardt on the AD Jacobian (∂_σ, ∂_{amp}, ∂_{θ_i})
$\sigma = 0.460 \pm 0.021$
amp $= 1.037 \pm 0.030$
$\chi^2 = 98.4$ for 118 dof
120 stars, $J = 0.02, 0.08, 0.2$
LM path (10 steps)
start
truth
best fit, 68/95% contours
amp
σ